SPEC'26 AI Problem: LLM-Based CNC Part Analysis

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Choose your level, build as far as you can, and demo it on hackathon day: 27 Oct 2026, 10:30 AM, NED University, Karachi. Teams of up to 4.

1. The challenge

CNC manufacturing information is scattered across Excel spreadsheets, 3D CAD models and PDF engineering drawings. Finding and combining it for one part by hand takes time and engineering expertise.

Problem statement. Build a lightweight, LLM-powered application that accepts a component's part number (for example, 2280) and automatically retrieves and analyzes its associated files to answer basic engineering and manufacturing questions.

The system should combine three sources and present a concise, evidence-based analysis of the selected part:

- An Excel-based manufacturing record
- An IGES (.igs / .iges) 3D model
- A PDF engineering drawing

Sample files for the example part are provided separately by the organizers.

2. Levels: choose how far to go

Every team picks a level and builds as far as it can. Each level includes the one before it. Reaching a higher level earns more credit, and a complete lower level beats an unfinished higher one.

Level	What your app must do
1. Core	Take a part number, look it up in the Excel manufacturing record, and use an LLM to answer basic questions about that part in plain language.

Level	What your app must do
2. Add the drawing	Also read the PDF engineering drawing and combine it with the Excel data. Every answer states which file it came from.
3. Add the 3D model	Also read the IGES model and use geometry facts you can extract reliably. Cross-check the three sources and flag anything that disagrees. Every answer cites its evidence.
Stretch	Work for any part number in the dataset, not only the example part.

An app that says "I can't find that in the files" is better than one that guesses.

3. Deliverables and submission

Submit before you present:

- **Working application or notebook** with a README that explains how to run it. Python is a common choice, but not required.
- **Technical document** covering your architecture, how each file type (Excel, PDF, IGES) is read, how the LLM is used, and the limitations of your approach.
- **Presentation** with a live demo for at least one part number, and the level you reached.

Teams may use any LLM they have access to, either a hosted API or a model running locally. Name the one you used in your document.

4. How you will be judged

Judges score every team on the same sheet. The first two criteria reward the level you reached.

Criterion	What judges look for	Weight (%)
Correct, evidence-based answers	Answers match the files and name their source	30

Criterion	What judges look for	Weight (%)
Sources combined	How many of the Excel, PDF and IGES sources are used, and how well they are cross-checked	25
Honesty and robustness	Says when information is missing instead of inventing it; handles odd questions	15
Engineering quality	Clean, runnable code with a clear README	15
Presentation and documentation	Clear demo, clear document, teamwork	15